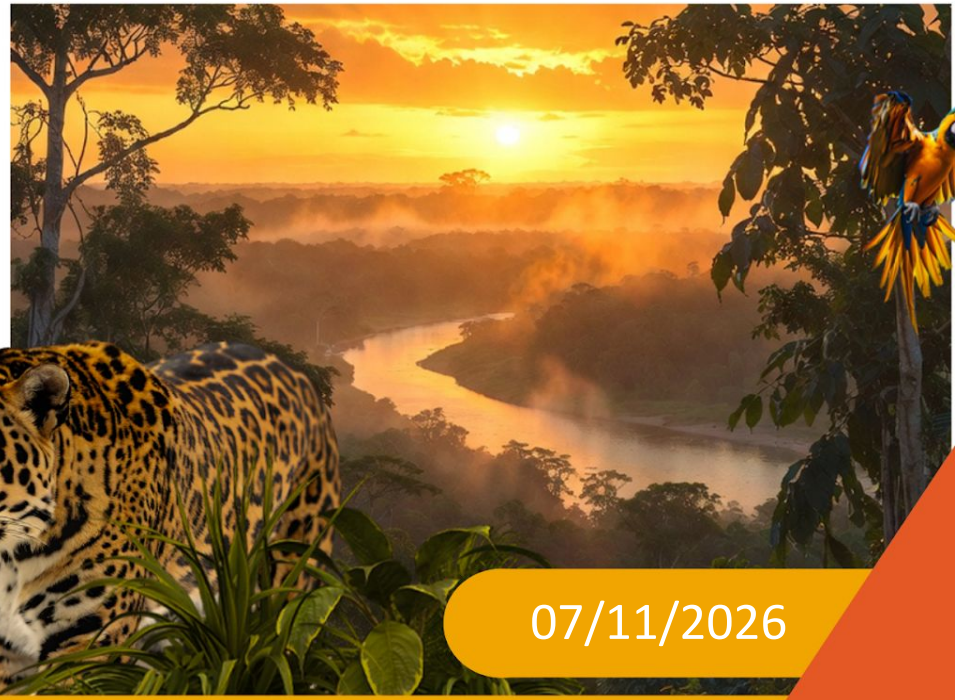
STAC e WTSS

Karine Reis Ferreira
Rennan Marujo
Gabriel Sansigolo
Felipe Menino Carlos

Brazilian National Institute for Space Research (INPE)



07/11/2026



SpatioTemporal Asset Catalog

A specification that defines a common structure, in JSON and GEOJSON files, for describing and cataloging geographic data.

Source: https://stacspec.org/en



SpatioTemporal Asset Catalog

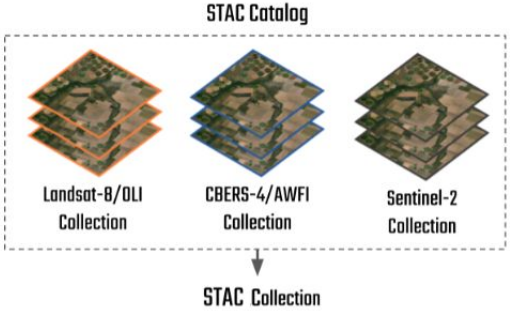
What is STAC?

A catalog
A Language for geospatial catalogs
Useful for search and discovery
Can be static or dynamic
Extensible

Source: https://stacspec.org/en

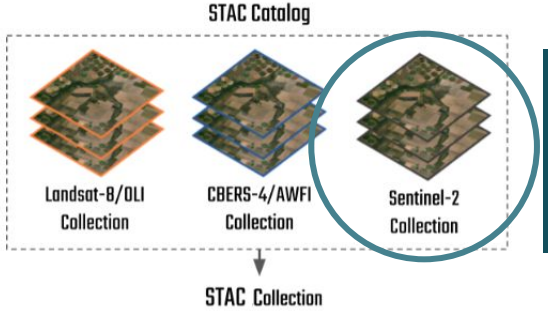
STAC

SpatioTemporal Asset Catalog



STAC

SpatioTemporal Asset Catalog

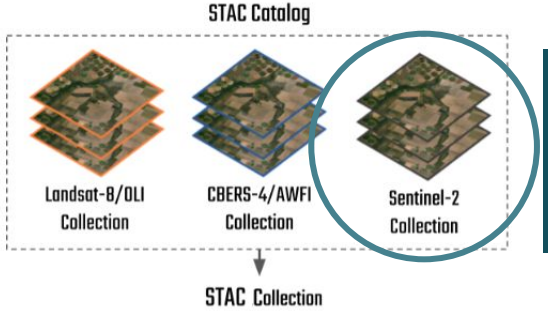


`Collection:` groups similar data, such as a set of images obtained from the same sensor or satellite.

STAC

SpatioTemporal Asset Catalog

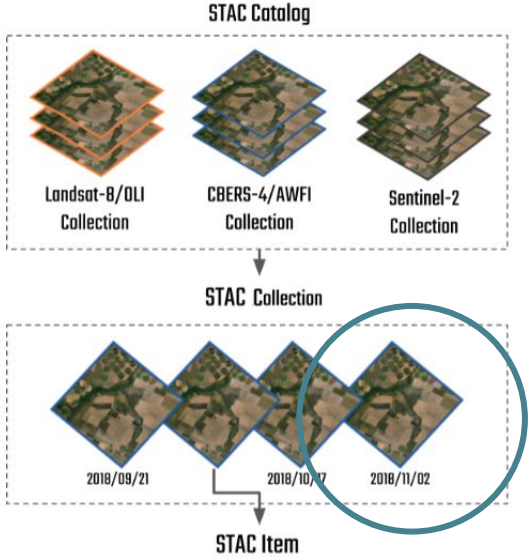
```
{
  "type": "Collection",
  "id": "CB4A-WFI-L4-SR",
  "license": "CC-BY-SA-4.0",
  "extent": {
    "spatial": { "bbox": [[-74, -34, -34, 6]] },
    "temporal": {
      "interval": [["2020-01-01T00:00:00Z", null]]
    }
  },
  "providers": [{ "name": "INPE", … }],
  "summaries": { … }
}
```



`Collection:` groups similar data, such as a set of images obtained from the same sensor or satellite.

STAC

SpatioTemporal Asset Catalog

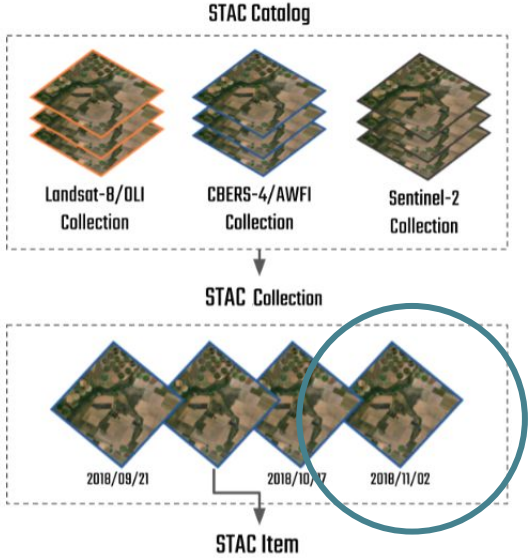


`Item:` represents a specific data unit, such as images corresponding to an exact location or moment, with spatial, temporal, and spectral metadata.

STAC

SpatioTemporal Asset Catalog

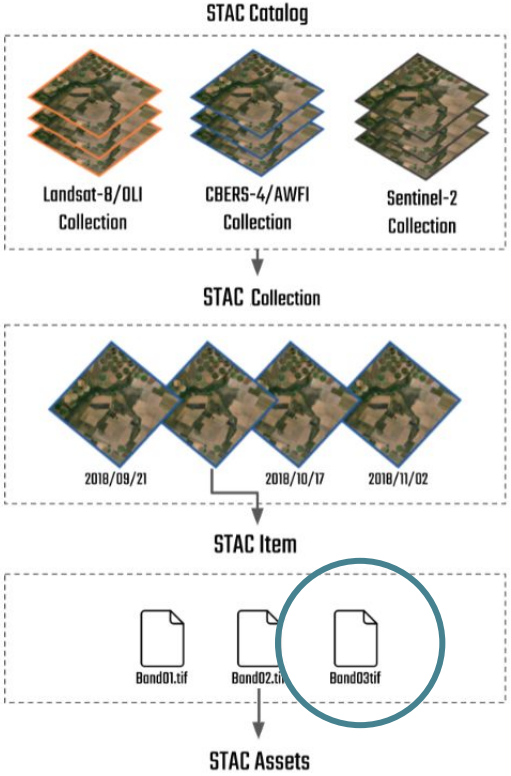
```
{
  "type": "Feature",
  "stac_version": "1.1.0",
  "id": "CBERS4A_WFI_214_132_20240512",
  "collection": "CB4A-WFI-L4-SR",
  "geometry": { "type": "Polygon", … },
  "bbox": [-48.1, -16.2, -46.9, -15.1],
  "properties": {
    "datetime": "2024-05-12T13:22:10Z",
    "eo:cloud_cover": 3.5
  },
  "links": [ … ],
  "assets": { … }
}
```



`Item:` represents a specific data unit, such as images corresponding to an exact location or moment, with spatial, temporal, and spectral metadata.

STAC

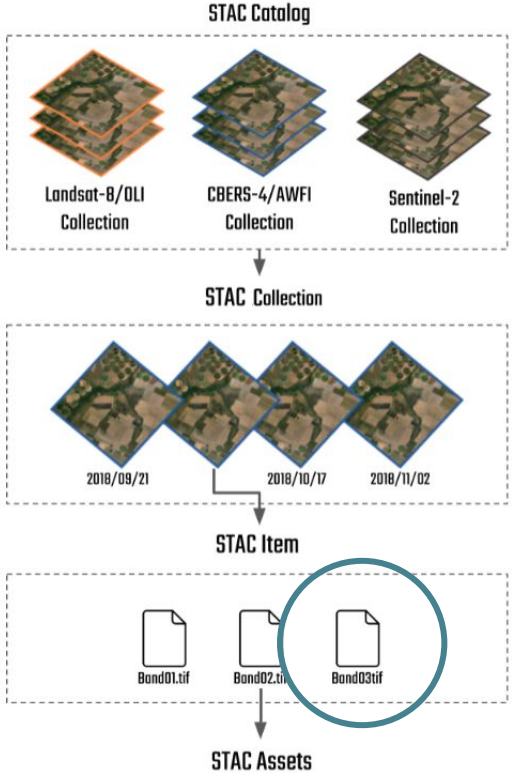
SpatioTemporal Asset Catalog



`Asset:` files associated with the Item, which may include different formats, resolutions, variables, auxiliary data, among other aspects.

STAC

SpatioTemporal Asset Catalog




```
"assets": {
  "BAND13": {
    "href": "https://…/BAND13.tif",
    "type": "image/tiff; application=geotiff",
    "roles": ["data"],
    "title": "Banda 13 (azul)"
  },
  "thumbnail": {
    "href": "https://…/thumbnail.png",
    "type": "image/png",
    "roles": ["thumbnail"]
  }
}
```

`Asset:` files associated with the Item, which may include different formats, resolutions, variables, auxiliary data, among other aspects.

STAC

SpatioTemporal Asset Catalog

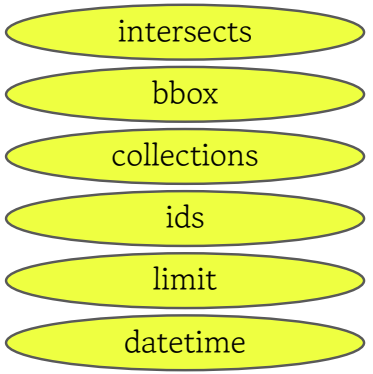
rel	Points to
self	The document itself
root	The root Catalog
paren	The level immediately above
child	A child Catalog or Collection
item	An item contained herein
collection	The Collection to which the item belongs

```
{
  "rel": "parent",
  "href": "../collection.json",
  "type": "application/json"
}
```

STAC

SpatioTemporal Asset Catalog

Endpoint	Returns
GET /	Home page (Root catalog)
GET / collections	List of collections
GET / collections / {id}	A collection
…/{id}/items	Items in a collection
GET | POST /search	Search across all collections



https://stac-extensions.github.io/

Extensions

Disasters Charter*	disaster	Collection, Item	Proposal	1.1.0	2024-06-05	Disaster Charter Extension Specification
Earthquake	eq	Collection, Item	Proposal	1.0.0	2025-01-20	Earthquake Extension Specification
Electro-Optical	eo	Collection, Item	Stable	2.0.0	2024-09-09	Covers electro-optical data that represents a snapshot of the Earth. It could consist of cloud cover and multiple spectral bands, for example visible bands, infrared bands, red edge bands and panchromatic bands.
Example Links	example	Catalog, Collection, Item	Proposal	0.0.1	2023-08-14	Allows to provide links to examples, e.g. code snippets.
File Info	file	Catalog, Collection, Item	Stable	2.1.0	2021-12-11	Specifies file-related details such as size, data type and checksum for assets and links in STAC.
Film*	film	Collection, Item	*WIP*	*Unknown*	*Unknown*	Film related metadata, such as roll, negative sequence and other physical attributes. Especially useful for digitised historic aerial imagery surveys.

Clients and public catalogs

- pystac-client
- rstac
- rustac

```python
from pystac_client import Client

catalogo = Client.open("https://data.inpe.br/bdc/stac/v1/")

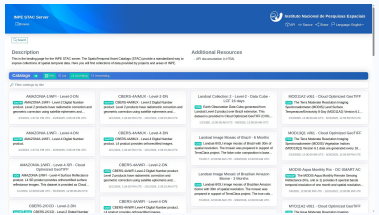
busca = catalogo.search(
    collections=["CB4A-WFI-L4-SR-1"],
    bbox=[-48.1, -16.2, -46.9, -15.1],
    datetime="2024-01-01/2024-06-30",
    max_items=5,
)

for item in busca.items():
    print(item.id, item.datetime, list(item.assets))
```

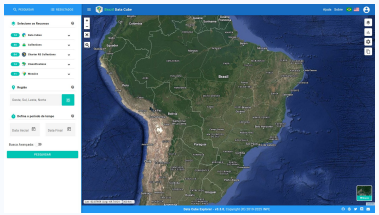
- INPE
- Microsoft Planetary Computer
- Earth Search (Element 84)
- USGS Landsat
- Copernicus Data Space

Integrated Catalog

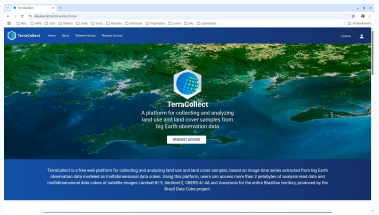
Aplications



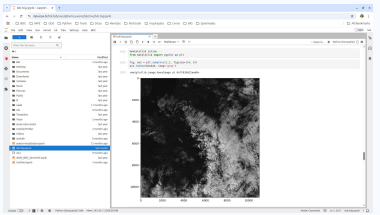
STAC Browser\



DataCube Explorer



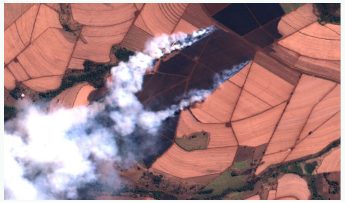
TerraCollect



Python & R

BDC - STAC

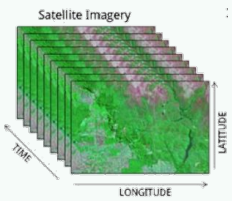
Data Collection



CBERS-4A/WPM



Sentinel-2



Cubos Dados



GOES

Outras coleções

INPE STAC Server

Browse

Instituto Nacional de Pesquisas Espaciais

API Source Share Language: English

Search

Description

This is the landing page for the INPE STAC server. The SpatioTemporal Asset Catalogs (STAC) provide a standardized way to expose collections of spatial temporal data. Here you will find collections of data provided by projects and areas of INPE.

Additional Resources

- API documentation in HTML

Catalogs 80

Tiles List Ascending Descending

Filter catalogs by title

AMAZONIA-1/WFI - Level-2-DN

GeoTiff AMAZONIA-1/WFI - Level-2 Digital Number product. Level 2 products have radiometric correction and geometric correction using satellite ephemeris and…

3/3/2021, 12:57:23 PM UTC - 4/21/2026, 3:50:22 PM UTC

CBERS-4/WFI 3M Mosaic

COG CBERS-4/WFI image mosaic of Brazil with 64m of spatial resolution. The mosaic was prepared in order to demonstrate the technological capabilities of the Brazil…

4/1/2020, 12:00:00 AM UTC - 6/30/2020, 12:00:00 AM UTC

GOES-19 Cloud & Moisture Imagery

NetCDF The GOES-19 Advanced Baseline Imager (ABI) L2 Cloud and Moisture Imagery provides 16 spectral bands with high temporal resolution over the American…

4/2/2025, 12:00:00 AM UTC - 4/22/2026, 3:40:00 PM UTC

MYD13Q1 v061 - Cloud Optimized GeoTIFF

COG The Aqua Moderate Resolution Imaging Spectroradiometer (MODIS) Vegetation Indices (MYD13Q1) Version 6.1 data are generated every 16…

7/4/2002, 12:00:00 AM UTC - 4/14/2026, 11:59:59 PM UTC

AMAZONIA-1/WFI - Level-4-DN

GeoTiff AMAZONIA-1/WFI - Level-4 Digital Number product.

3/17/2021, 12:00:00 AM UTC - 4/21/2026, 2:18:42 PM UTC

CBERS-4A/MUX - Level-2-DN

GeoTiff CBERS-4A/MUX - Level-2 Digital Number product. Level 2 products have radiometric correction and geometric correction using satellite ephemeris and…

12/27/2019, 1:53:00 PM UTC - 4/18/2026, 3:04:34 PM UTC

Landsat 3M Amazon Biome Mosaic

COG Landsat-8/OLI image mosaic of Brazilian Amazon biome with 30m of spatial resolution. The mosaic was prepared in support of TerraClass project. The true colo…

7/1/2016, 12:00:00 AM UTC - 9/30/2016, 12:00:00 AM UTC

S2 1M Amazon Biome Mosaic

COG Sentinel-2 image mosaic of Brazilian Amazon biome with 10m of spatial resolution. The mosaic was prepared to support BiomasBR Program to produce Th…

10/1/2020, 12:00:00 AM UTC - 12/31/2020, 12:00:00 AM UTC

AMAZONIA-1/WFI - Level-4-SR - Cloud Optimized GeoTIFF

COG AMAZONIA-1/WFI - Level-4 Surface Reflectance product. L4 SR product provides orthorectified surface reflectance images. This dataset is provided as Cloud…

1/1/2024, 12:00:00 AM UTC - 4/9/2026, 12:00:00 AM UTC

CBERS-4A/MUX - Level-4-DN

GeoTiff CBERS-4A/MUX - Level-4 Digital Number product. L4 product provides orthorectified images.

12/27/2019, 1:53:00 PM UTC - 4/18/2026, 3:03:29 PM UTC

Landsat 6M Mosaic

COG Landsat-8/OLI image mosaic of Brazil with 30m of spatial resolution. Th… TerraClass project.

7/1/2017, 12:00:00

S2 2M Cerrado Biome Mosaic

COG Sentinel-2 image mosaic of Brazilian Cerrado

data.inpe.br/stac/browser

AMAZONIA-1/WFI Image Mosaic of Brazil - 3 Months

CBERS-4A/MUX - Level-4-SR

COG CBERS-4A/MUX - Level-4 Surface Reflectance product over Brazil and part of South America. L4 SR

Landsat 6M São Paulo State Mosaic

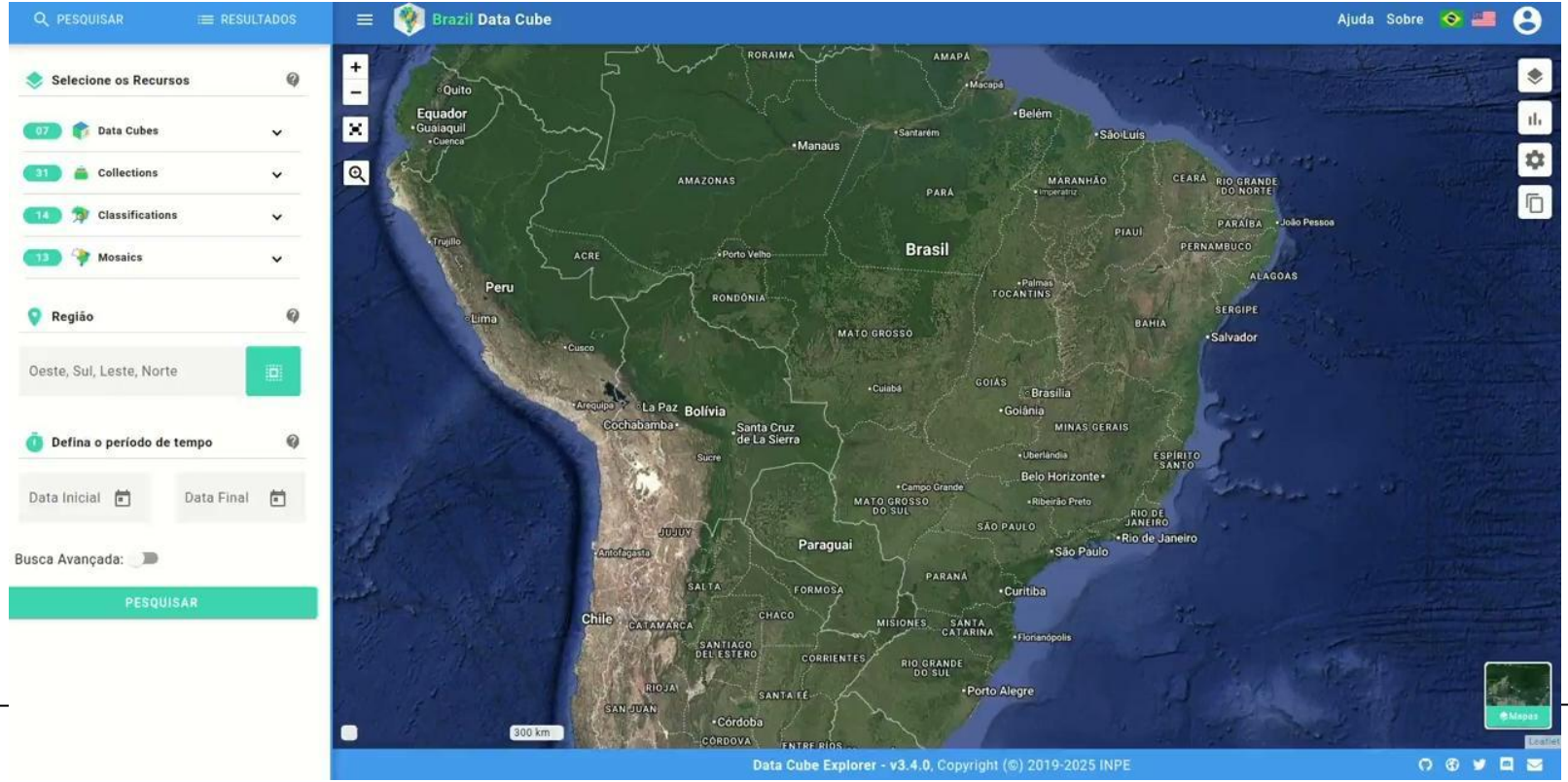
COG Earth Observation Data Cube generated from

S2 3M B08B04B03 Amazon Biome Mosaic

COG Sentinel-2 image mosaic of Brazilian Amazon

Data Cube Explorer: search and temporal visualization

https://data.inpe.br/bdc/explorer



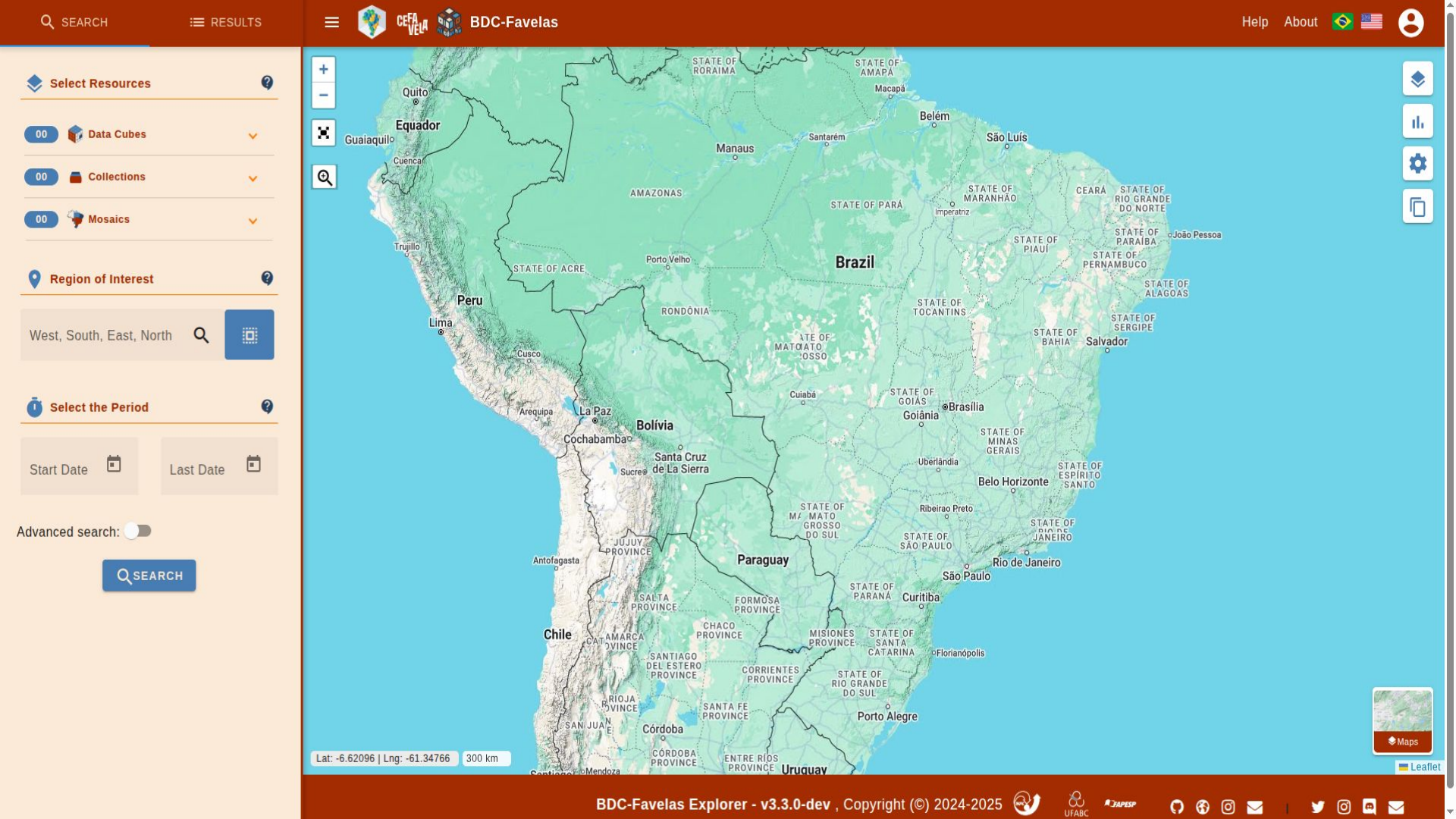
BDC-Favelas

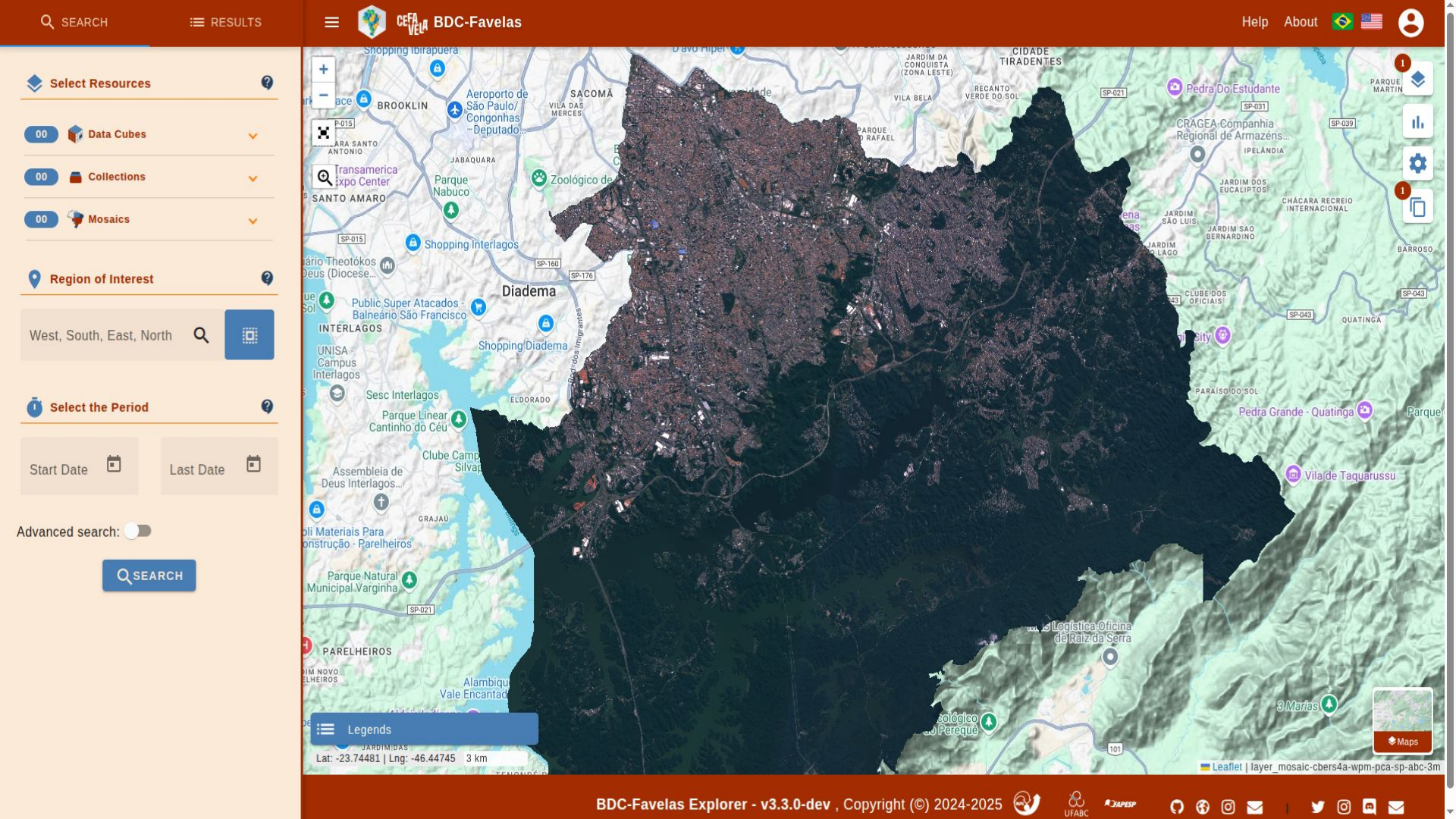
National Technology in the Service of Popular Territories

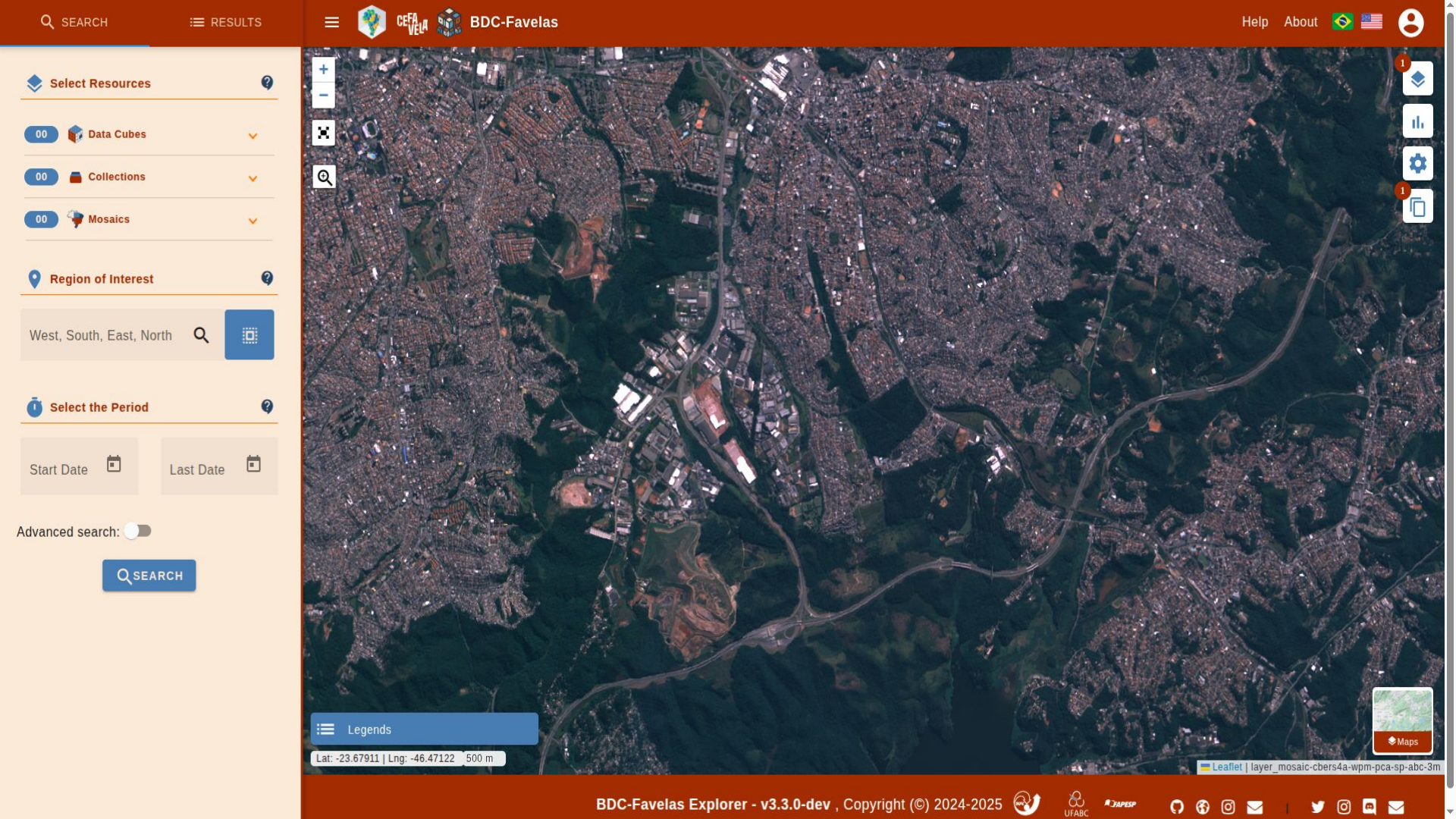


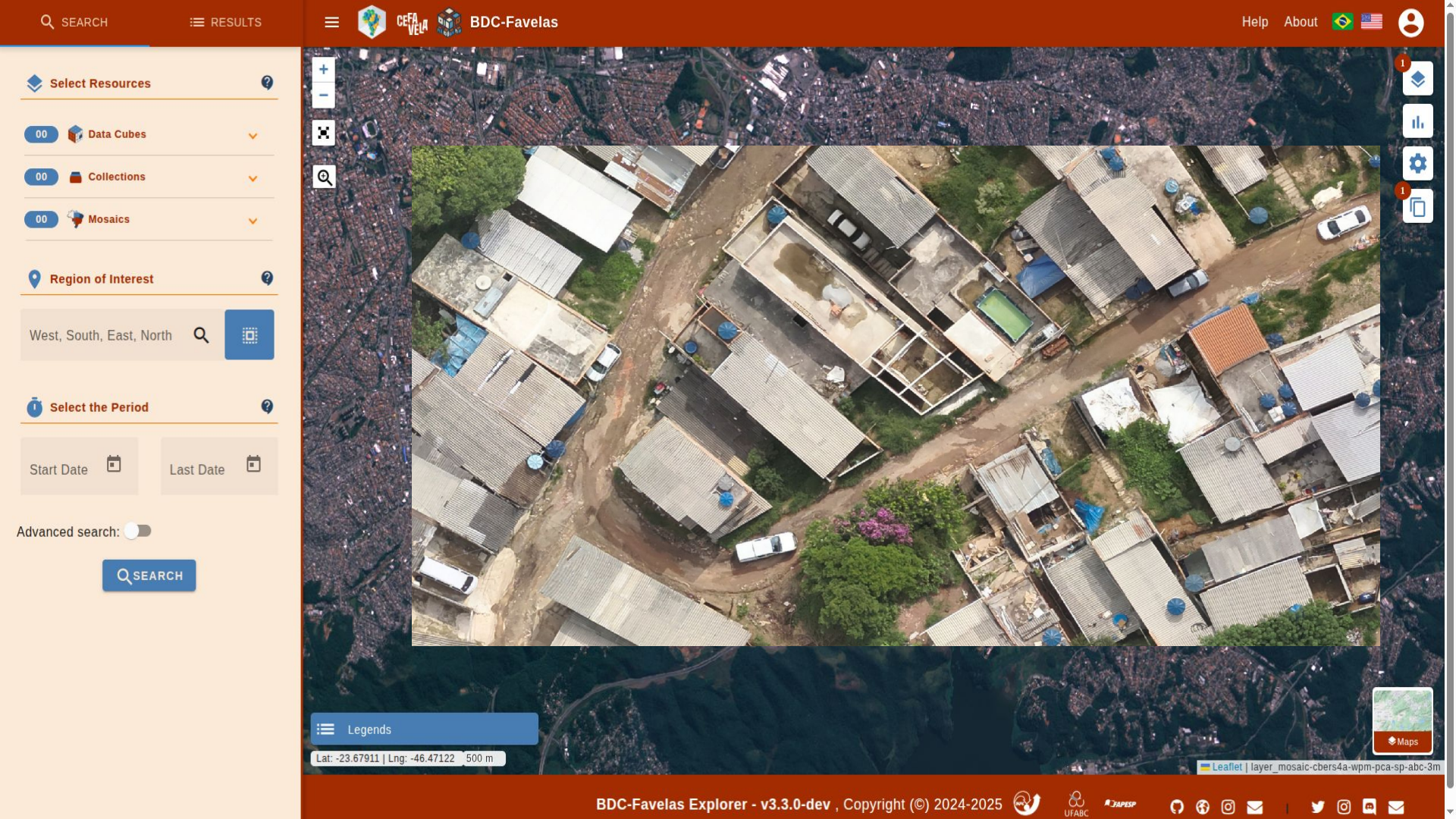
A **data cube** of products derived from satellite and drone imagery for Brazilian favelas

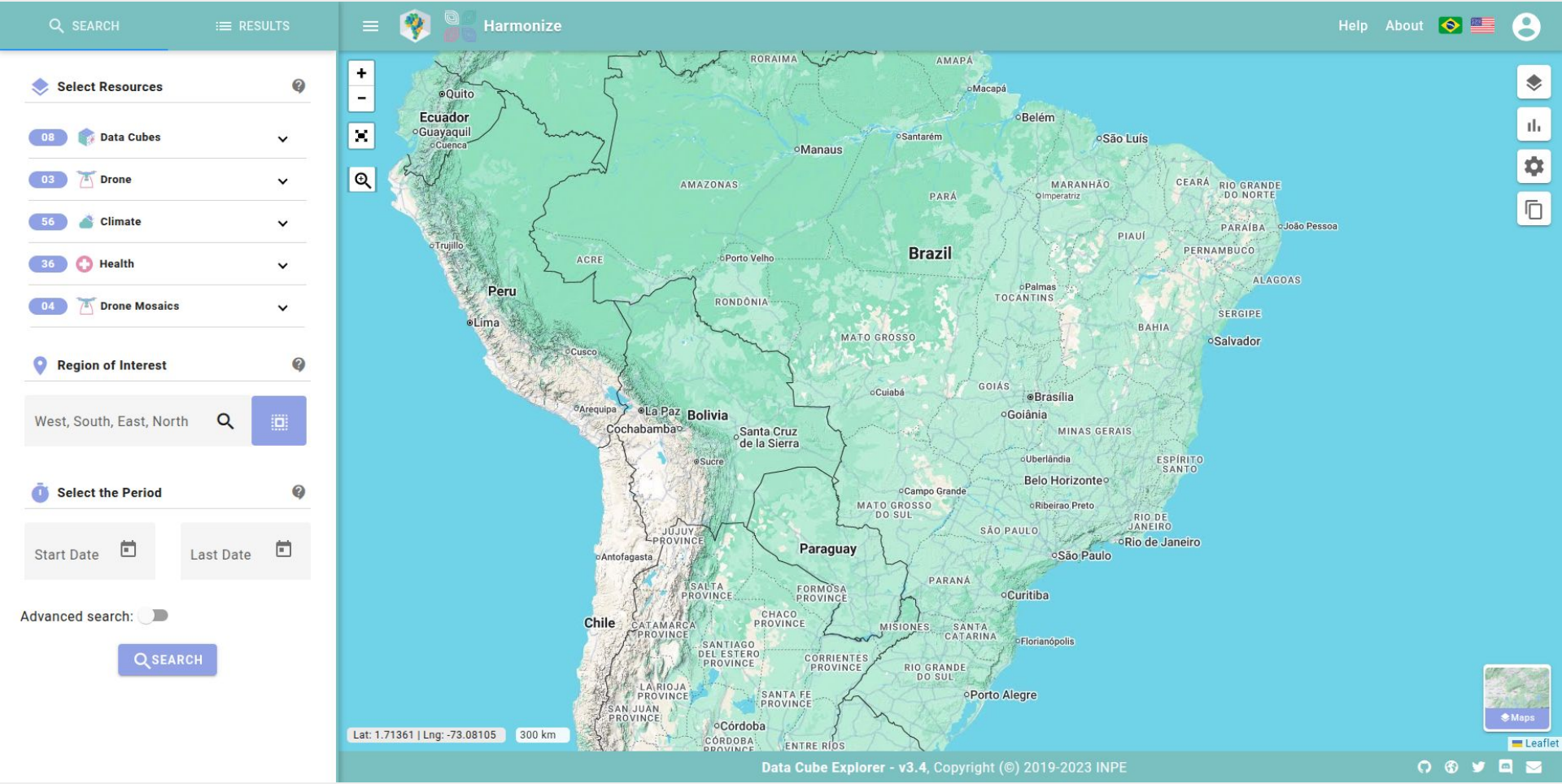
Public, free, and open access via a web portal, with data collections ready for analysis


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BDC-Favelas
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Guaiaquil
Cuenca
Manaus
Santarém
Belém
São Luís
Macapá
AMAZONAS
STATE OF PARÁ
Imperatriz
CEARA
STATE OF MARANHÃO
STATE OF RIO GRANDE DO NORTE
STATE OF PARAÍBA
João Pessoa
STATE OF PIAUÍ
STATE OF PERNAMBUCO
STATE OF ALAGOAS
STATE OF SERGIPE
Salvador
STATE OF BAHIA
Trujillo
STATE OF ACRE
Porto Velho
Brazil
Peru
Lima
RONDÔNIA
STATE OF TOCANTINS
Cusco
Cuiabá
STATE OF GOIÁS
Goiânia
Brasília
Arequipa
La Paz
Bolívia
Cochabamba
Santa Cruz de La Sierra
Sucre
Uberlândia
STATE OF MINAS GERAIS
Belo Horizonte
STATE OF ESPÍRITO SANTO
Ribeirao Preto
STATE OF MATO GROSSO DO SUL
STATE OF SÃO PAULO
Rio de Janeiro
São Paulo
Antofagasta
JUJUY PROVINCE
Paraguay
SALTA PROVINCE
FORMOSA PROVINCE
STATE OF PARANÁ
Curitiba
Chile
CHACO PROVINCE
MISIONES PROVINCE
STATE OF SANTA CATARINA
Florianópolis
SANTIAGO DEL ESTERO PROVINCE
CORRIENTES PROVINCE
STATE OF RIO GRANDE DO SUL
Porto Alegre
SANTA FE PROVINCE
Córdoba
CÓRDOBA PROVINCE
ENTRE RIOS PROVINCE
Uruguay
Lat: -6.62096 | Lng: -61.34766
300 km
Maps
Leaflet
BDC-Favelas Explorer - v3.3.0-dev , Copyright (©) 2024-2025


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SEARCH
Legends
Lat: -23.74481 | Lng: -46.44745
3 km
Leaflet | layer_mosaic-cbers4a-wpm-pca-sp-abc-3m
Maps
Diadema
INTERLAGOS
SANTO AMARO
BROOKLIN
JABAQUARA
SACOMÃ
CIDADE TIRADENTES
ELDORADO
GRAJAU
PARELHEIROS
Shopping Ibirapuera
Aeroporto de São Paulo/ Congonhas - Deputado...
Parque Nabuco
Shopping Interlagos
Public Super Atacados - Balneário São Francisco
Shopping Diadema
UNISA - Campus Interlagos
Sesc Interlagos
Parque Linear Cantinho do Céu
Assembleia de Deus Interlagos...
Parque Natural Municipal Varginha
Pedra Do Estudante
CRAGEA-Companhia Regional de Armazéns...
Pedra Grande - Quatinga
Vila de Taquarussu
Logística Oficina de Raiz da Serra
3 Marias
BDC-Favelas Explorer - v3.3.0-dev , Copyright (©) 2024-2025
UFABC
FAPESP


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Lat: -23.67911 | Lng: -46.47122
500 m
Maps
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BDC-Favelas Explorer - v3.3.0-dev , Copyright (©) 2024-2025


SEARCH
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Select the Period
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Last Date
Advanced search:
SEARCH
Legends
Lat: -23.67911 | Lng: -46.47122
500 m
Maps
Leaflet | layer_mosaic-cbers4a-wpm-pca-sp-abc-3m
BDC-Favelas Explorer - v3.3.0-dev , Copyright (©) 2024-2025
UFABC


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04 Drone Mosaics
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SEARCH
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Ecuador
Guayaquil
Cuenca
Manaus
Macapá
Belém
Santarém
São Luís
Imperatriz
Trujillo
Peru
Lima
Cusco
Arequipa
Porto Velho
Brazil
Palmas
João Pessoa
Salvador
Cuiabá
Brasília
Goiânia
La Paz
Bolivia
Cochabamba
Santa Cruz de la Sierra
Sucre
Uberlândia
Belo Horizonte
Campo Grande
Ribeirao Preto
Rio de Janeiro
São Paulo
Paraguay
Antofagasta
Chile
Curitiba
Florianópolis
Porto Alegre
Córdoba
Lat: 1.71361 | Lng: -73.08105
300 km
Maps
Leaflet
Data Cube Explorer - v3.4, Copyright (©) 2019-2023 INPE

SpatioTemporal Asset Catalog

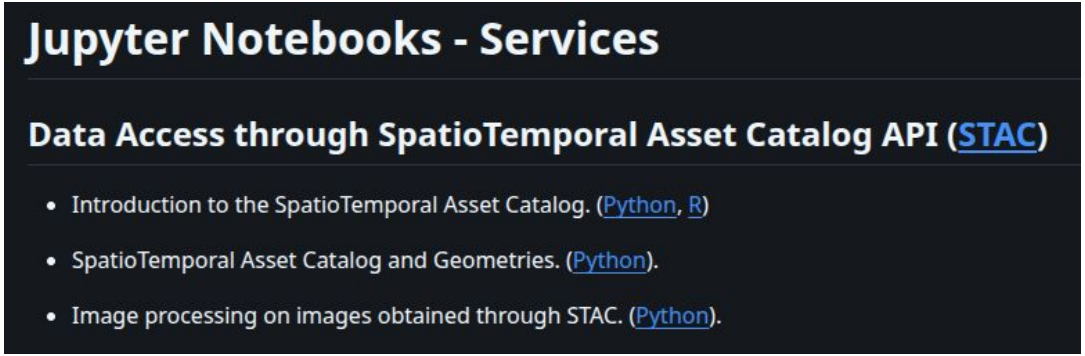
INPE STAC: https://data.inpe.br/bdc/stac/v1/

INPE STAC Browser: https://data.inpe.br/stac/browser/?.language=en

BDC Explorer: https://data.inpe.br/bdc/explorer/explore

Code-gallery: https://github.com/brazil-data-cube/code-gallery





Source: https://stacspec.org/en

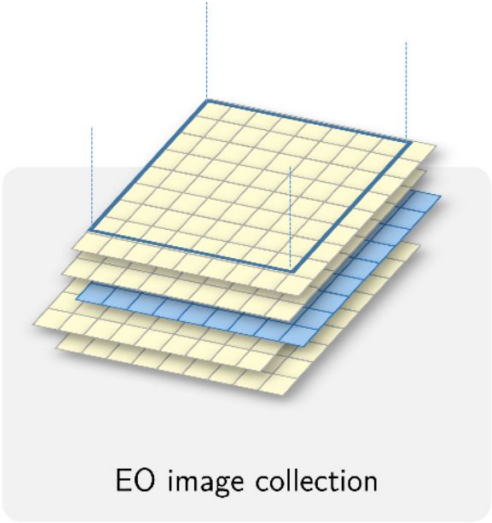


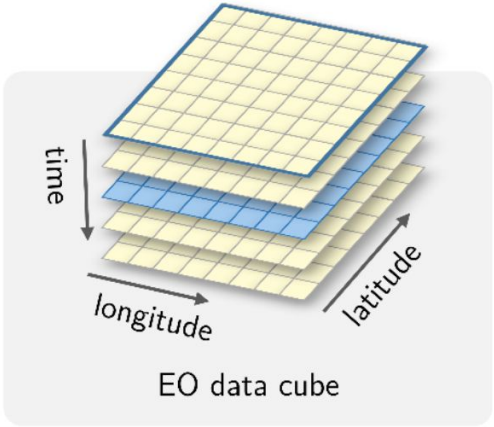
Web Time Series Service
(WTSS)

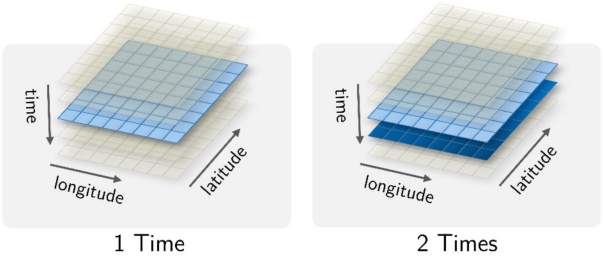
Simple API
Bridging the gap between remote sensing applications and time series data.

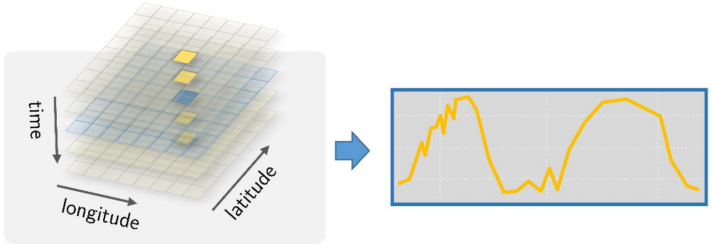
Source: https://stacspec.org/en

WTSS

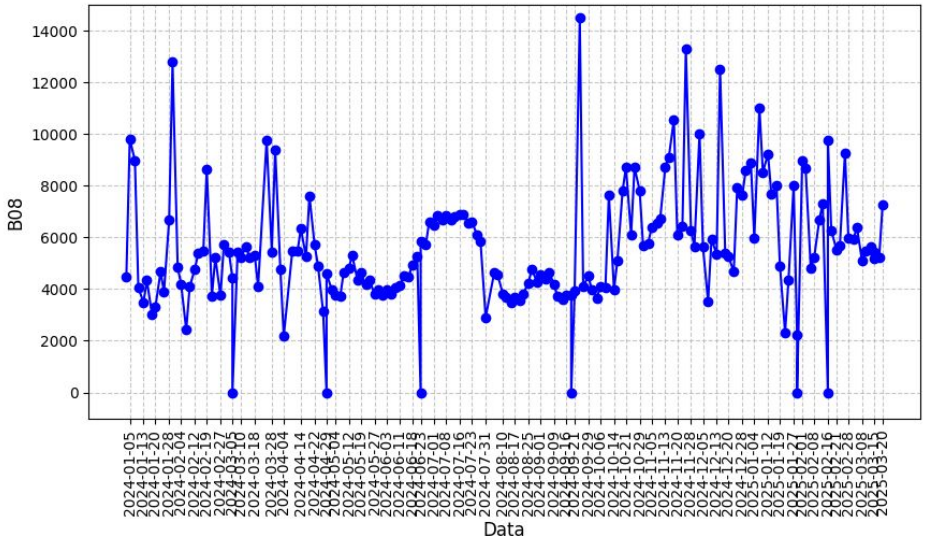

EO image collection


time
longitude
latitude
EO data cube

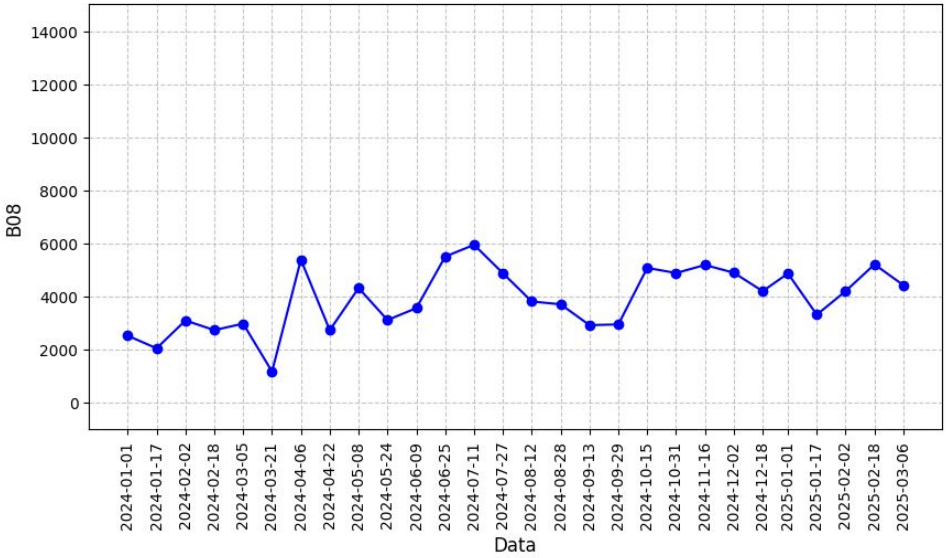

time
longitude
latitude
1 Time
time
longitude
latitude
2 Times
Image selection


time
longitude
latitude
Time series extraction

Original collection time series



Data cube time series



WTSS

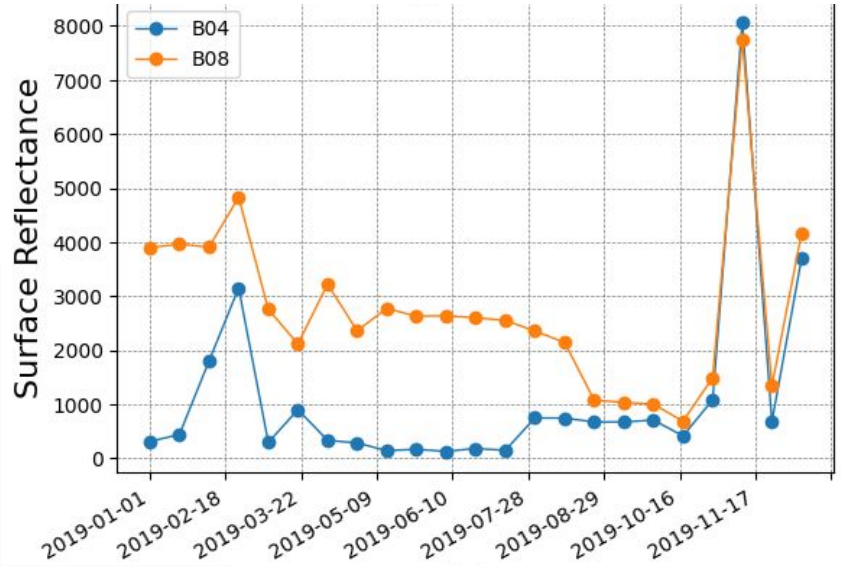
```
from wtss import *

service = WTSS('https://data.inpe.br/bdc/wtss/v4/')

coverage = service['S2-16D-2']

ts = coverage.ts(attributes=('B04', 'B08'),
                 latitude=-9.41866, longitude=-61.46103,
                 start_date='2019-01-01', end_date='2019-12-31')

print(ts.B04)
print(ts.B08)
```



WTSS

Anais XVII Simpósio Brasileiro de Sensoriamento Remoto - SBSR, João Pessoa-PB, Brasil, 25 a 29 de abril de 2015, INPE

WTSS: um serviço web para extração de séries temporais de imagens de sensoriamento remoto

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Abstract. This paper presents a lightweight web service named Web Time Series Service (WTSS) for handling

Brazilian Journal of Cartography (2017), Nº 69/5, Special Issue GEOINFO 2017: 913-922
Brazilian Society of Cartography, Geodesy, Photgrammetry and Remote Sense
ISSN: 1808-0936

WEB SERVICES FOR BIG EARTH OBSERVATION DATA

Serviços Web para Grandes Dados de Observação da Terra

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Received on January 18, 2017/ Accepted on April 7, 2017
Recebido em 18 de Janeiro, 2017/ Aceito em 7 de Abril, 2017

ABSTRACT

The aim of geospatial web services is to enable users to share and process geospatial data on the Web, no matter what platform or protocol is used. In this paper, we investigate what the design decisions needed to implement web services

Service Operations:

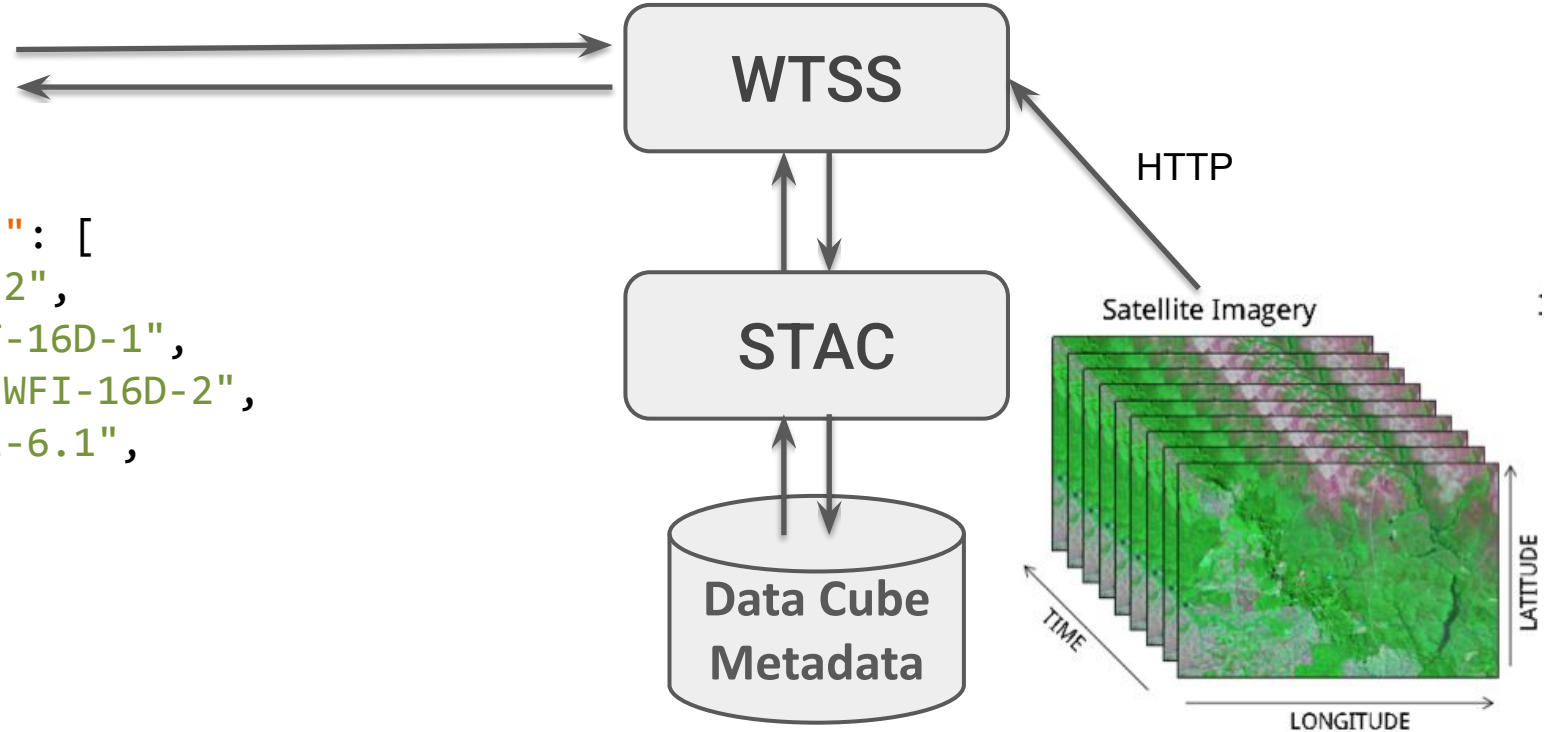
List Coverages
Describe Coverage
Time Series

WTSS: list_coverages



/wtss/**list_coverages**

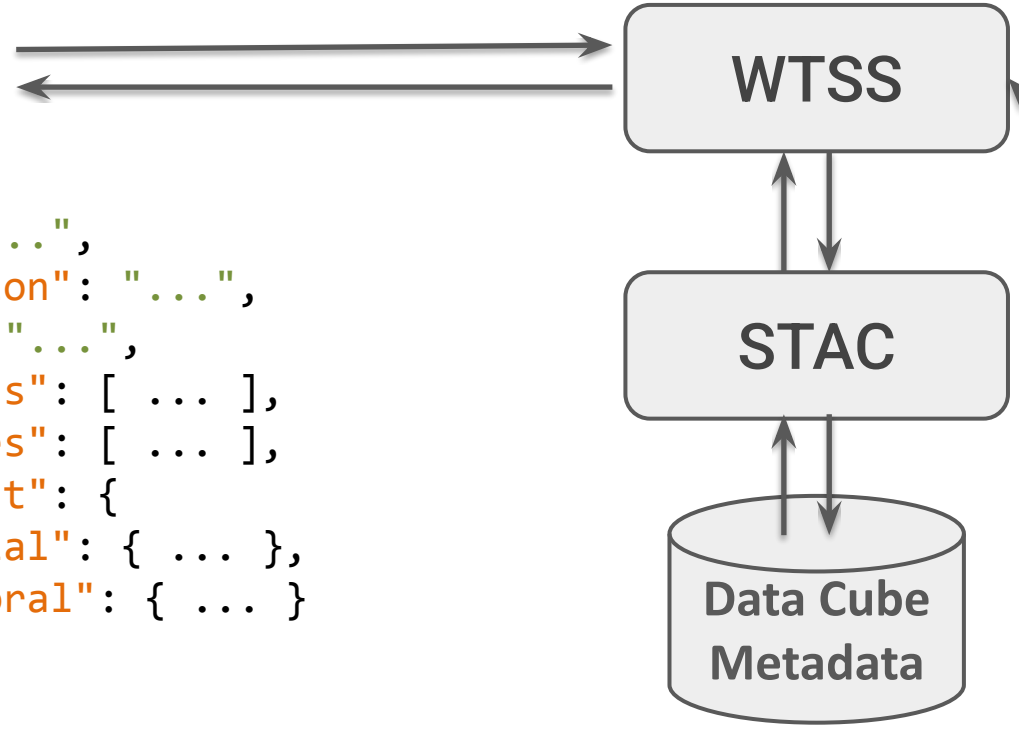
```
{
  "coverages": [
    "S2-16D-2",
    "LANDSAT-16D-1",
    "CBERS4-WFI-16D-2",
    "myd13q1-6.1",
    ...
  ]
}
```



WTSS: describe_coverage



/wtss/**describe_coverage**




```
{
  "name": "...",
  "description": "...",
  "detail": "...",
  "dimensions": [ ... ],
  "attributes": [ ... ],
  "geo_extent": {
      "spatial": { ... },
      "temporal": { ... }
  }, ...
}
```

WTSS: time_series

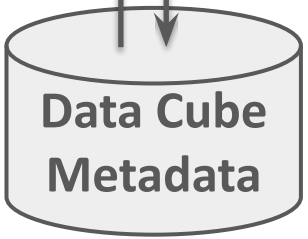


/wtss/**time_series**

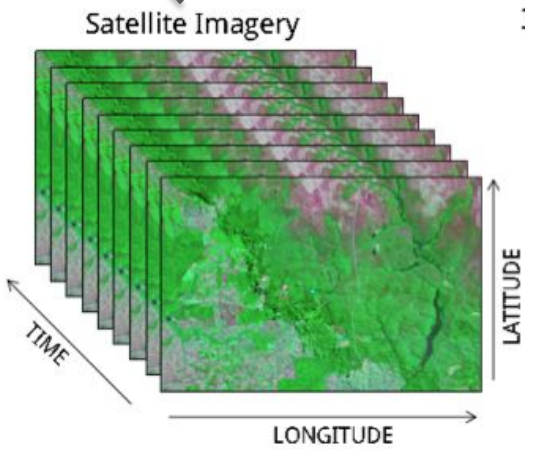




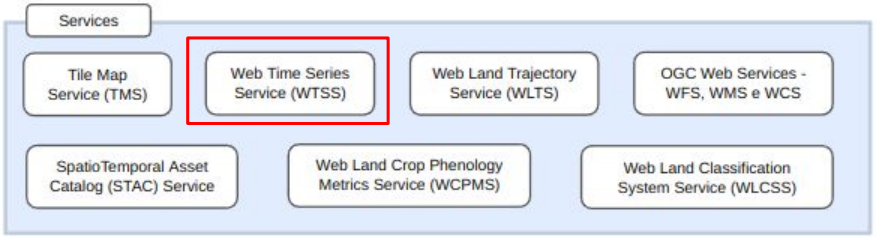








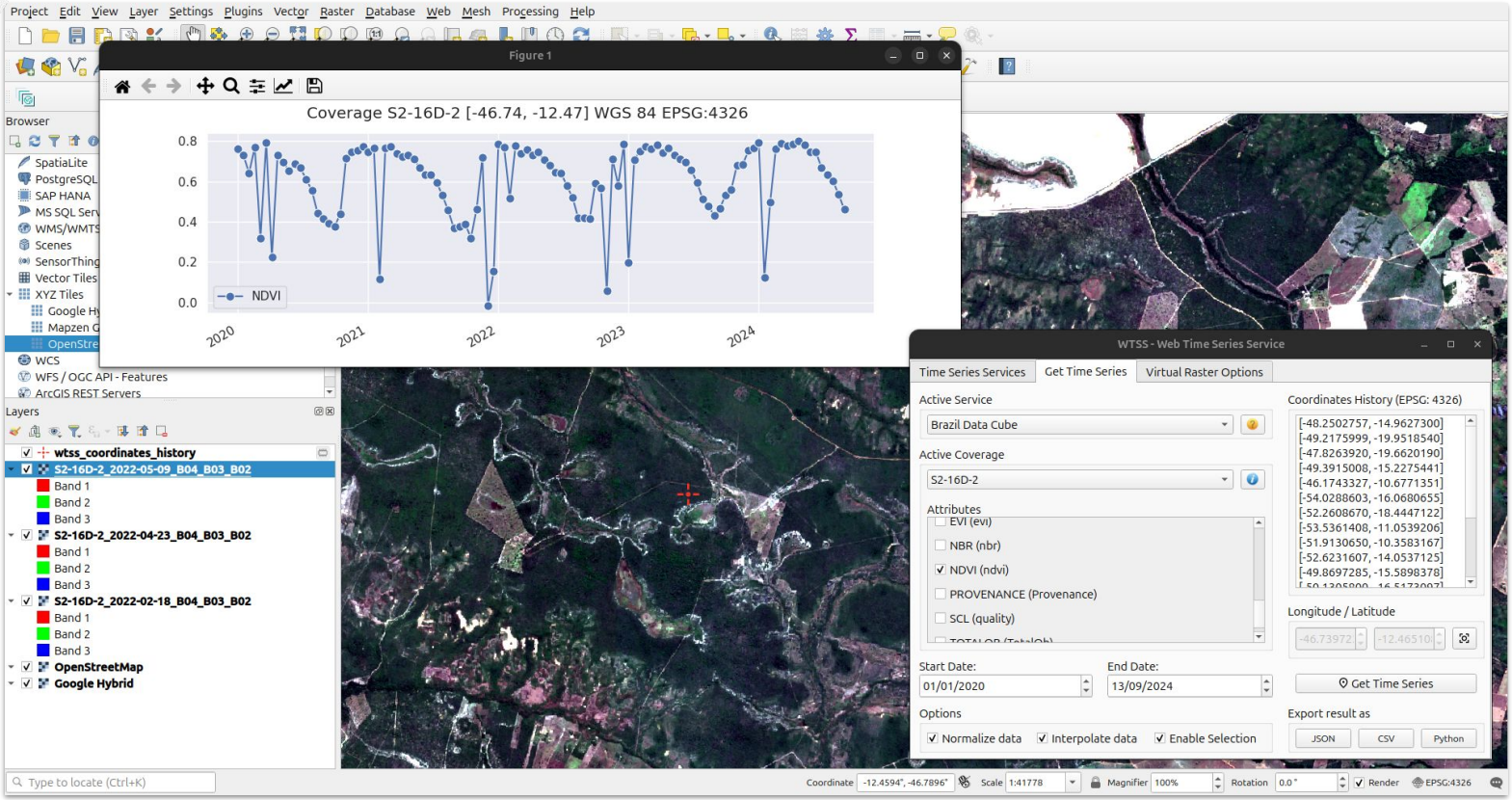

```
{
  "result": {
    "attributes": [ ... ],
    "timeline": [ ... ]
  },
  "query": { ... }
}
```



Web Time Series Service (WTSS)



Plugin QGIS

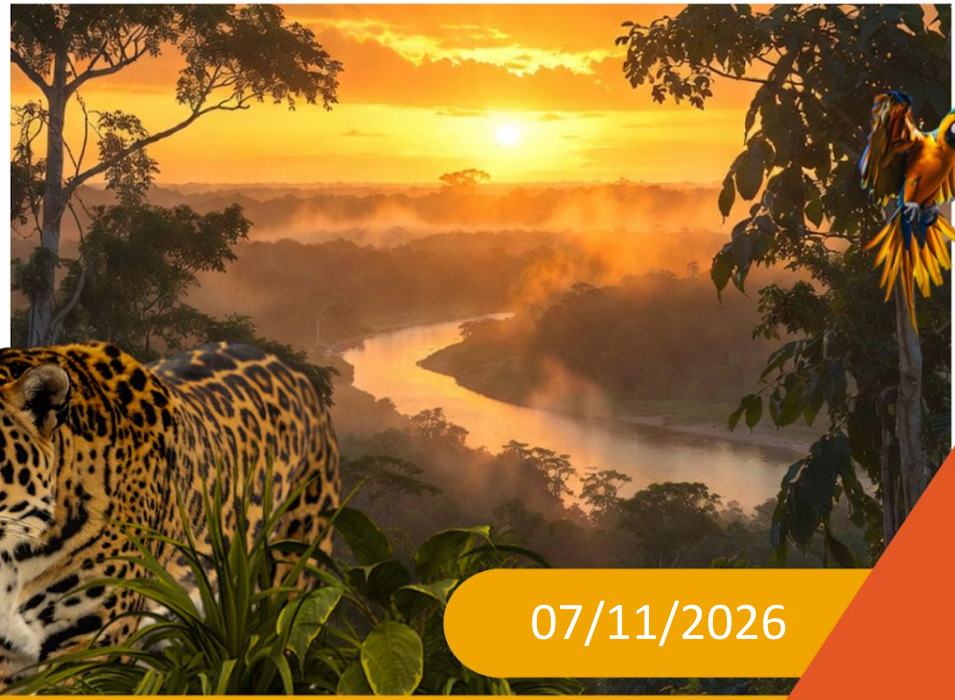


https://github.com/brazil-data-cube/wtss-qgis

STAC e WTSS

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07/11/2026